

MANISTEE COUNTY BLACKER AP, MI, USA

WMO: 726385

Lat: 44.267N

Lon: 86.250W

Elev: 189

StdP: 99.07

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 94894

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.0	-13.9	-21.1	0.6	-15.1	-18.6	0.7	-13.0	11.3	-6.3	10.5	-6.3	3.9	90	0.564

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	29.5	22.2	27.9	20.9	27.0	20.2	23.4	28.0	22.3	26.6	21.3	25.2	4.4	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.0	17.1	26.5	21.0	16.1	25.2	19.9	14.9	24.2	70.3	28.0	66.4	26.4	62.7	25.0	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.0	8.7	7.8	-20.9	32.1	3.7	1.4	-23.5	33.1	-25.6	33.9	-27.7	34.7	-30.3	35.7		
			-21.4	24.9	3.6	1.2	-24.0	25.7	-26.1	26.4	-28.1	27.0	-30.7	27.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.6	-4.7	-4.1	-0.2	6.0	12.2	17.1	19.6	18.9	15.4	9.1	3.1	-1.7
	DBStd	9.82	5.16	5.32	5.96	5.01	4.72	4.04	3.38	3.11	4.23	4.53	4.57	4.62
	HDD10.0	1998	456	396	323	141	31	2	0	0	5	72	211	363
	HDD18.3	4085	714	629	575	369	199	69	27	31	108	287	456	621
	CDD10.0	1132	0	0	7	22	98	216	296	277	166	46	6	1
	CDD18.3	177	0	0	1	1	8	33	64	49	19	2	0	0
	CDH23.3	1499	0	0	4	9	95	314	558	365	141	13	0	0
	CDH26.7	317	0	0	1	1	19	74	130	68	24	1	0	0
Wind	WSAvg	3.4	4.1	4.0	3.7	3.9	3.4	2.8	2.6	2.6	2.9	3.5	3.9	3.9
Precipitation	PrecAvg	876	55	42	50	75	86	87	84	82	89	89	76	64
	PrecMax	1154	111	80	125	136	207	216	173	142	170	227	175	191
	PrecMin	684	16	18	9	20	21	30	18	22	14	18	18	12
	PrecStd	116	26	18	26	30	44	52	40	29	40	50	40	34
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.2	12.2	22.0	24.8	28.9	31.1	32.1	31.1	29.6	25.2	18.8	13.0
		MCWB	7.0	8.8	14.8	15.9	20.5	22.9	24.0	23.1	22.5	18.9	13.9	10.7
	2%	DB	7.0	8.0	16.2	21.3	26.2	28.8	29.4	28.3	27.1	22.2	15.2	8.8
		MCWB	4.6	5.0	11.1	13.4	18.0	21.1	22.5	21.8	20.2	16.8	11.2	7.0
	5%	DB	3.9	5.3	12.2	17.7	23.2	27.2	27.8	27.1	25.0	19.7	12.8	7.0
		MCWB	2.2	2.8	8.2	10.9	16.3	19.7	21.3	20.8	19.3	15.4	9.4	4.8
	10%	DB	2.5	2.9	8.0	14.8	21.2	25.0	26.4	25.4	22.8	17.2	11.0	4.2
		MCWB	1.2	1.0	4.9	8.9	14.7	18.5	20.1	19.8	18.0	13.4	8.1	2.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.0	9.0	15.7	17.4	21.5	24.0	25.1	24.2	23.0	20.5	14.5	11.6
		MCDB	8.6	10.1	20.8	22.8	27.4	29.5	30.4	28.2	27.6	23.5	17.5	13.0
	2%	WB	4.6	5.5	11.9	14.5	19.4	22.0	23.7	22.9	21.5	18.1	12.3	7.5
		MCDB	6.4	8.3	15.3	19.8	24.6	27.5	28.3	27.0	25.1	20.7	14.9	8.5
	5%	WB	2.7	3.0	8.1	11.9	17.7	20.6	22.4	21.9	20.2	15.6	10.0	4.7
		MCDB	3.9	5.0	11.0	16.6	22.1	25.3	26.6	25.5	23.7	19.2	12.7	6.7
	10%	WB	1.1	1.3	5.3	9.6	15.7	19.5	21.3	20.8	19.0	13.6	7.9	2.8
		MCDB	2.5	3.1	8.4	14.5	19.9	23.7	25.2	24.1	22.4	16.7	10.3	4.6
Mean Daily Temperature Range	5% DB	MDBR	6.0	7.4	9.1	10.5	11.8	11.4	11.2	10.9	11.2	9.7	7.3	5.7
		MCDBR	8.1	9.4	13.0	15.1	14.5	13.4	12.7	12.3	12.9	13.1	11.5	8.8
		MCWBR	7.0	7.4	9.2	9.1	8.5	7.3	6.9	6.9	7.5	8.6	8.7	7.6
	5% WB	MCDBR	7.5	8.6	12.0	13.6	12.6	11.4	10.8	10.1	10.7	11.2	10.6	8.4
		MCWBR	6.8	7.2	9.1	9.3	8.4	7.0	6.4	6.2	6.9	8.4	8.6	7.6
Clear-Sky Solar Irradiance	taub	0.292	0.307	0.329	0.360	0.390	0.404	0.405	0.397	0.379	0.346	0.314	0.286	
	taud	2.434	2.373	2.403	2.348	2.334	2.334	2.345	2.361	2.400	2.475	2.501	2.484	
	Ebn at Noon	836	883	908	902	881	867	861	856	843	829	804	810	
	Edh at Noon	73	92	102	118	123	124	121	115	102	83	68	63	
All-Sky Solar Radiation	RadAvg	1.09	1.99	3.51	4.67	5.67	6.21	6.30	5.34	4.17	2.50	1.38	0.86	
	RadStd	0.12	0.22	0.38	0.44	0.49	0.42	0.28	0.34	0.37	0.29	0.20	0.11	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page